Online Appendix

Anticipation, Not Reaction: Media Discourse, FOMC Communication, and Pre-Announcement Volatility

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This appendix documents the corpus coverage in full, the remaining identification output, the computational architecture, and the reproducibility package. Section and equation references of the form "main text" point to the manuscript.

A Corpus coverage by lag

Table 1 reports the number of meetings with media coverage at each lag for the three corpora, extending Table 1 of the main text. The broad corpus has complete coverage at every lag from -3 to $+3$ for the 33 meetings in the estimation window; the curated outlets have uneven coverage that thins at the extreme lags. This table is the reproducible basis for the sample restrictions used throughout the paper: every reported n can be read off it.

Table 1: Meetings with coverage by source and lag.

Lag k	GDELT	Guardian	NYT
-7		18	6
-6		19	8
-5		22	13
-4		15	6
-3	33	5	10
-2	33	13	12
-1	33	15	12
+0	33	62	21
+1	33	37	22
+2	33	29	22
+3	33	14	8
+4		14	6
+5		12	6
+6		23	7
+7		20	14

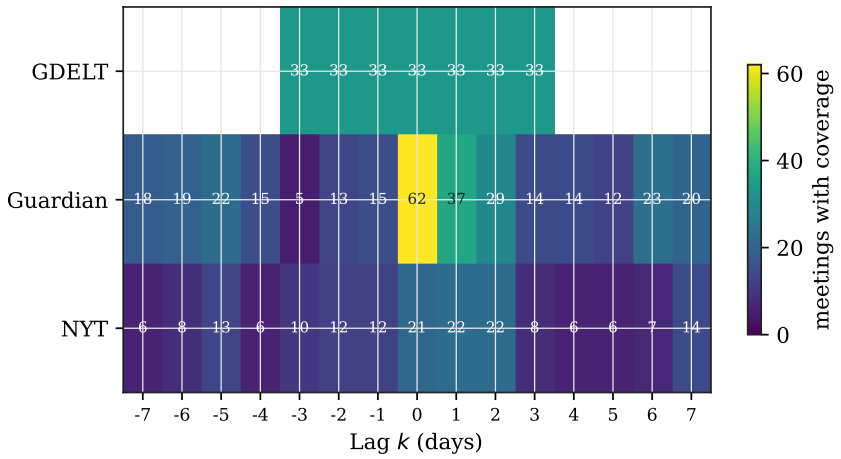


Figure 1: Coverage heatmap. Darker cells indicate more meetings covered at that source and lag.

B Additional identification output

B.1 Granger causality

Table 2 gives the full Granger output for the test of whether pre-meeting similarity precedes post-meeting volatility across the chronologically ordered meetings. The evidence of temporal precedence is marginal, significant at the second lag and not at the first, which is why the main text treats the Granger result as descriptive rather than as evidence of a causal direction.

Table 2: Granger causality, sim_{-2} to post-meeting volatility.

Lag	F	df	p-value
1	1.244	(1,29)	0.2739
2	3.510	(2,26)	0.0447
3	2.425	(3,23)	0.0915

B.2 Vector autoregression

Table 3 reports the impulse response of standardized post-meeting volatility to a one-standard-deviation shock in standardized pre-meeting similarity, from a reduced-form VAR(2) on similarity, volatility, and the VIX. The cumulative response is positive but modest, consistent with the controlled regression in which the similarity coefficient is small once the VIX is included.

Table 3: VAR(2) impulse response of volatility to a similarity shock.

Horizon (meetings)	Vol. response to sim. shock
0	0.000
1	0.005
2	0.497
3	0.464
4	0.293
5	0.203
6	0.161

B.3 Scatter of the raw relationship

Figure 2 plots the bivariate relationship behind column (1) of the main regression table.

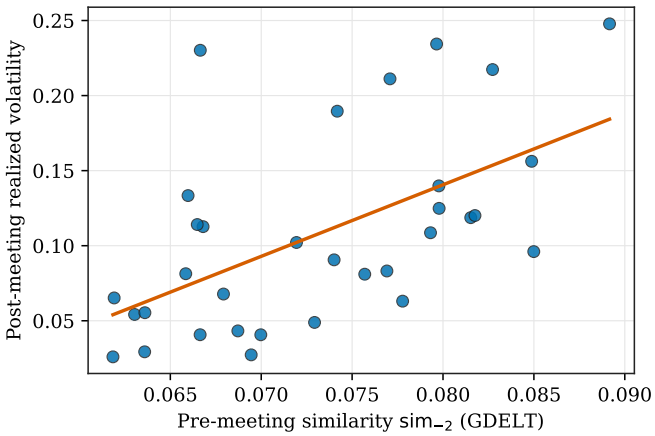


Figure 2: Pre-meeting similarity against post-meeting realized volatility, broad corpus, with the ordinary-least-squares fit.

C Tone measure and timeline

Figure 3 shows the net hawkishness of the FOMC minutes across the sample, recovered entirely from the transparent lexicon. The measure reproduces the known turning points of the decade without any hand labeling, which is the evidence that the tone signal is meaningful before any market variable is introduced.

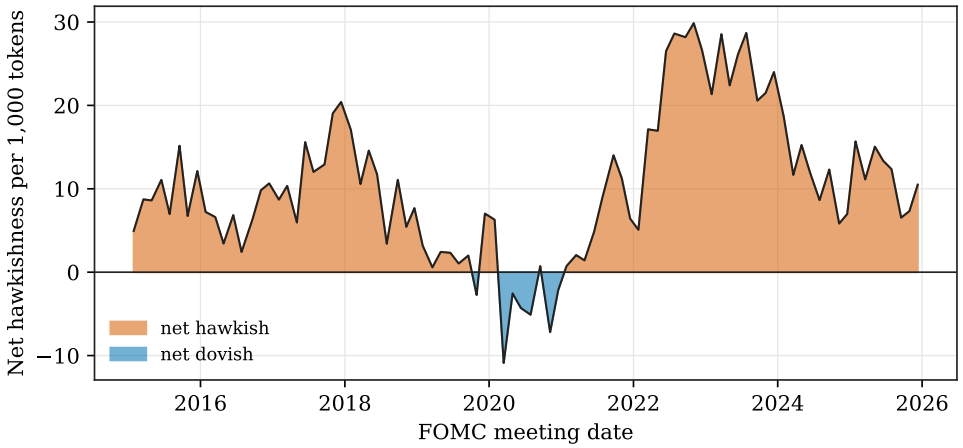


Figure 3: Net hawkishness per 1,000 tokens across the 88 FOMC minutes, 2015–2025.

D Computational architecture

The pipeline separates resource-gated stages from the reproducible CPU path.

Distributed stages. GDELT ingestion reads the Global Knowledge Graph shards, filters to United States Federal-Reserve-relevant records with Spark SQL, and writes date-partitioned

Parquet (`scripts/ingest/06_spark_ingest_gdelt.py`). Tone scoring broadcasts the hawkish–dovish lexicon to the executors and applies a grouped pandas user-defined function (`scripts/sentiment/07_spark_sentiment.py`). Cosine similarity broadcasts the small per-meeting FOMC vectors and keeps the large media side distributed (`scripts/similarity/11_spark_similarity.py`). These stages run unchanged in a local Spark session for validation on a sample week and on a five-node cluster for the full corpus.

Resource-gated robustness. Contextual-embedding similarity with Sentence-BERT (`scripts/similarity/11b_sbert_similarity.py`) and FinBERT tone (`scripts/sentiment/07c_finbert_inference.py`) are implemented and run on a GPU node; over the full media corpus they are GPU-bound and are therefore provided as code rather than executed in the CPU package.

Reproducible CPU path. The panel construction, main regression, identification battery, tone validation, benchmark, tables, and figures run on a single node from the committed intermediate artifacts. The workflow is a Snakemake file with a pinned conda environment and a container image, so the command `snakemake -c4 all` regenerates every number and figure in the paper.

Throughput. The single-node scoring kernel is benchmarked in the main text (Table 7): the Numba parallel just-in-time implementation is roughly fifteen times faster than a pure-Python loop on a five-million-token workload, which is what makes full-corpus scoring feasible before the work is distributed.

E File-by-file map

Table 4: Pipeline scripts.

Stage	Script
Panel construction and coverage	`scripts/analysis/00_build_panel.py`
Main regression and lead–lag profile	`scripts/analysis/12_main_regression.py`
Placebo, Granger, VAR, bootstrap, subsamples	`scripts/analysis/13_identification.py`
LaTeX tables	`scripts/analysis/16_make_tables.py`
Loughran–McDonald tone validation	`scripts/sentiment/07b_lm_dictionary.py`
Distributed GDELT ingestion	`scripts/ingest/06_spark_ingest_gdelt.py`
Distributed tone scoring	`scripts/sentiment/07_spark_sentiment.py`
Distributed cosine similarity	`scripts/similarity/11_spark_similarity.py`
Sentence-BERT similarity (GPU)	`scripts/similarity/11b_sbert_similarity.py`
FinBERT tone (GPU)	`scripts/sentiment/07c_finbert_inference.py`
Throughput benchmark	`scripts/benchmark/13_benchmark.py`
Figures	`scripts/figures/make_figures.py`